Ashwin Vignesh M

AI Engineering · Computer Vision · LLM & GenAI · Data Science

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PROFILE

Master's student in AI Engineering at THI Ingolstadt, with hands-on experience building LLM-powered agents, full-stack RAG systems, and computer-vision pipelines through coursework and self-directed projects. Previously an ERP Analyst at DXC Technology, where I worked on Python automation, reporting, and dashboards. Looking for roles in AI engineering, computer vision, data science, and applied LLM / GenAI.

SKILLS

AI & Machine Learning:	Deep Learning, Computer Vision, LLMs, RAG, Generative AI, Reinforcement Learning, Statistical Analysis
Frameworks & Libraries:	PyTorch, Keras, TensorFlow, OpenCV, YOLOv8, MediaPipe, NumPy, Pandas, Scikit-learn, LangChain
Computer Vision:	Object Detection, Pose Estimation, Sensor Fusion (Camera–LiDAR), Synthetic Data, GANs
NLP & Generative AI:	Prompt Engineering, Agentic Workflows, Document Understanding, Semantic Retrieval, LLM API Integration (Claude, Gemini, OpenAI)
Programming:	Python, C/C++ (Basic), SQL (Basic), REST APIs
Web & Backend:	React, FastAPI, PostgreSQL, pgvector, ChromaDB, Redis, Streamlit, Docker
Tools & Systems:	Linux, Git, GitHub Actions, VS Code, Power BI, Power Automate, N8N, Jira, Simulink, SUMO
Languages:	English (Fluent), German A2 (B1 in progress)

WORK EXPERIENCE

Analyst II — ERP Package Applications
Sep 2022 – Mar 2024

DXC Technology, India

- Automated recurring data processing and reporting workflows in Python to support KPI-driven performance analysis.
- Built reusable scripts and internal documentation so automation jobs ran reliably across teams and environments.
- Created Power BI dashboards and technical reports for system monitoring, validation, and incident troubleshooting.
- Managed incident tickets via ServiceNow and provided day-to-day technical support for key client AT&T.

EDUCATION

M.Sc. AI Engineering for Autonomous Systems
Mar 2024 – Present

Technische Hochschule Ingolstadt, Germany

Coursework: Machine Learning, Deep Learning, Data Engineering, Sensor Data & Fusion (automotive focus), Advanced Mathematics for AI.

B.Tech in Mechanical Engineering
Aug 2018 – Jul 2022

Reva University, Bengaluru, India

Final Project: Automated Hydroponic Plant Grow System using Arduino & OpenCV. Published at NCNTME-2022.

PROJECTS

Agentic RAG Document Intelligence Platform

FastAPI · PostgreSQL (pgvector) · Gemini API · React · Docker · LangChain

- Built a full-stack RAG system for multi-document semantic search and structured Q&A with streaming responses.
- Added adaptive retrieval, cross-encoder reranking, and a production deployment on Render + Vercel with a responsive React UI.

- Implemented Graph RAG via Neo4j, plus conversation history, a user feedback loop, and an admin analytics dashboard.

JARVIS — Voice-Controlled AI Terminal Agent

Python · FastAPI · React · OpenAI Whisper · Claude API · ChromaDB · Docker

- Built a voice-controlled AI agent using OpenAI Whisper for speech recognition and Claude API for command execution, streaming results to a real-time React dashboard in under 3 seconds.
- Full-stack system with FastAPI, PostgreSQL, Redis, and ChromaDB RAG, deployed with Docker, NGINX, and GitHub Actions CI/CD.

Foundation Models for Computer Vision — Camera–LiDAR Fusion

Python · PyTorch · Jetson Nano · Git · KITTI / Waymo Datasets

- Led the project and set up GPU-based ML environments on Jetson Nano for foundation-model experimentation.
- Analyzed internal representations of foundation models for camera–LiDAR perception across large-scale datasets.
- Kept experiments reproducible via Git and delivered full technical documentation including final report and presentation.

Radar-AI: Object Detection with Synthetic Sensor Data

Python · GANs · CNNs · PyTorch · RADAR Signal Processing

- Developed GAN-based simulation pipelines to generate high-fidelity synthetic sensor data for rare driving scenarios.
- Evaluated synthetic data quality and measured its impact on downstream object detection performance.

Cricket Bat Swing Analyzer — Computer Vision

Python · YOLOv8 · MediaPipe · OpenCV · Streamlit

- Built an end-to-end CV pipeline using YOLOv8 for bat detection and MediaPipe for pose keypoint extraction from video.
- Computed physics-based swing metrics: peak speed, angular velocity, duration, angle, and a smoothness score.

Mini-CNN Framework from Scratch

C/C++ · CMake · LeNet-5 · Int8 Quantization · HPC

- Wrote a custom LeNet-5 inference engine in C/C++ without any ML framework, optimized with Im2Col for cache locality.
- Simulated Int8 quantization and integer arithmetic targeting edge hardware deployment.

Interactive Portfolio with AI Recruiter Assistant

React · Vite · Framer Motion · Gemini API · Tailwind CSS

- Built a personal portfolio with animated project visuals and an AI-powered technical match analysis tool.
- Designed structured LLM prompts that generate fit reports comparing candidate skills against job descriptions.

CERTIFICATIONS

Claude Code in Action	Anthropic	2026
Building Code Agents with Hugging Face	Hugging Face / DeepLearning.AI	2025
Introduction to Transformer-Based NLP	NVIDIA / Coursera	2024
Intro to Deep Learning	Kaggle	2024
Prompt Engineering for ChatGPT	Vanderbilt University	2023